EE1D1: Digital Systems A

BSc. EE, year 1, 2026-2027, lecture 1

Introduction to Digital Systems

Computer Engineering Lab

Faculty of Electrical Engineering, Mathematics & Computer Science

TU Delft

Delft University of Technology

Outline

Introduction

- Instructors
- Importance of Digital Systems
- Course Overview

Digital Systems

- Digital versus analog
- Binary Systems and Boolean Algebra
- Data representation
- Summary

Instructors: Arjan van Genderen

- Education
  - 1985 MSc. Electrical Engineering, TU Delft
  - 1991 PhD, TU Delft

- Lecturer for
  - Digital Systems A
  - Digital Systems B
  - IP-2 Smart Robot Challenge
  - Chip Design

- MSc. Coordinator for
  - Master Computer and Embedded Systems Engineering



A.J.vanGenderen@tudelft.nl

Room HB 09.080
Computer Engineering,
Faculty EEMCS, TU Delft

Instructors: Mansureh Shahraki Moghaddam

- Education
  - 2015: PhD from Indian Institute of Technology Delhi

- Lecturer:
  - Digital Systems A
  - Digital Systems B
  - IP-2 Smart Robot Challenge
  - Chip Design
  - Computer Architecture and Organization



M.ShahrakiMoghaddam@tudelft.nl
Room HB 09.080
Computer Engineering,
Faculty EEMCS, TU Delft

Instructors: Ton Slats

- EE Lab staff member:
  - Digital Systems A
  - Digital Systems B
  - IP-2 Smart Robot Challenge
  - Computer Architecture and Organization



A.M.J.Slats@TUDelft.nl

LB 01.260

Electrical Eng. Education
Faculty EEMCS, TU Delft

Importance of Digital Systems

- Digital Systems are everywhere.
- Examples
  - Smartphone
  - PC, game console
  - Communication
  - Industry
  - Household
  - Transport
  - Health
  - …









Digitals systems = Hardware (HW) + Software (SW)

Computer scientists: SW (+ HW) (e.g. databases, compilers)
Electrical engineers: HW+SW (e.g. instruments, chips)

Importance of Digital Systems

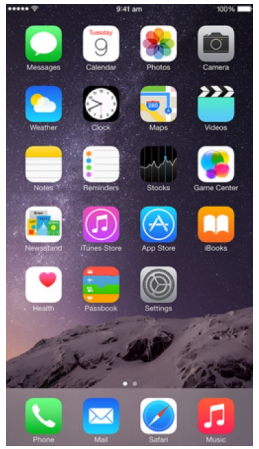
- iPhone 16





Hardware

+



Software

- apps
- ios
- drivers

Importance of Digital Systems

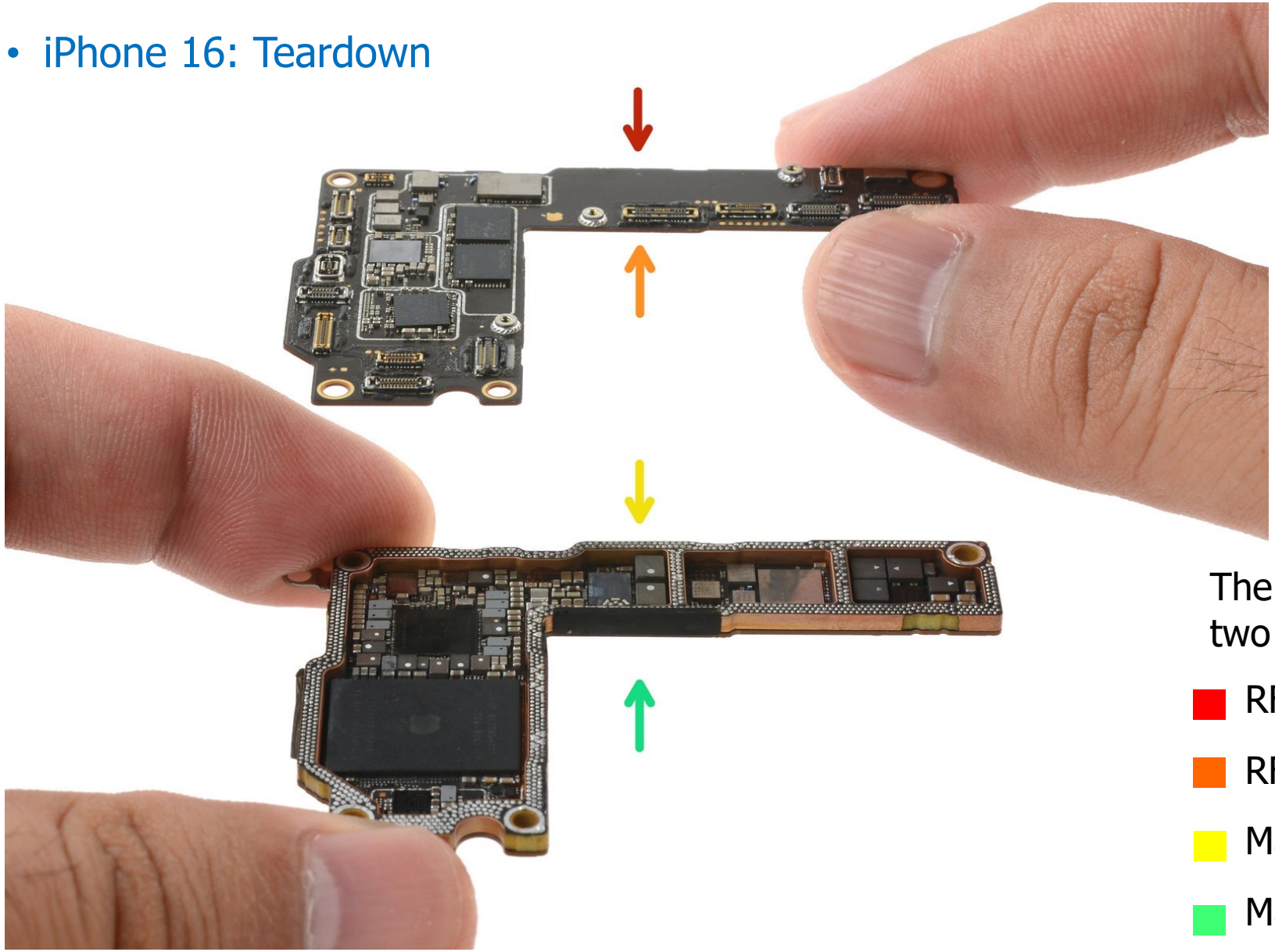
- iPhone 16 Teardown



source: www.how-fixit.com

Importance of Digital Systems

- iPhone 16: Teardown



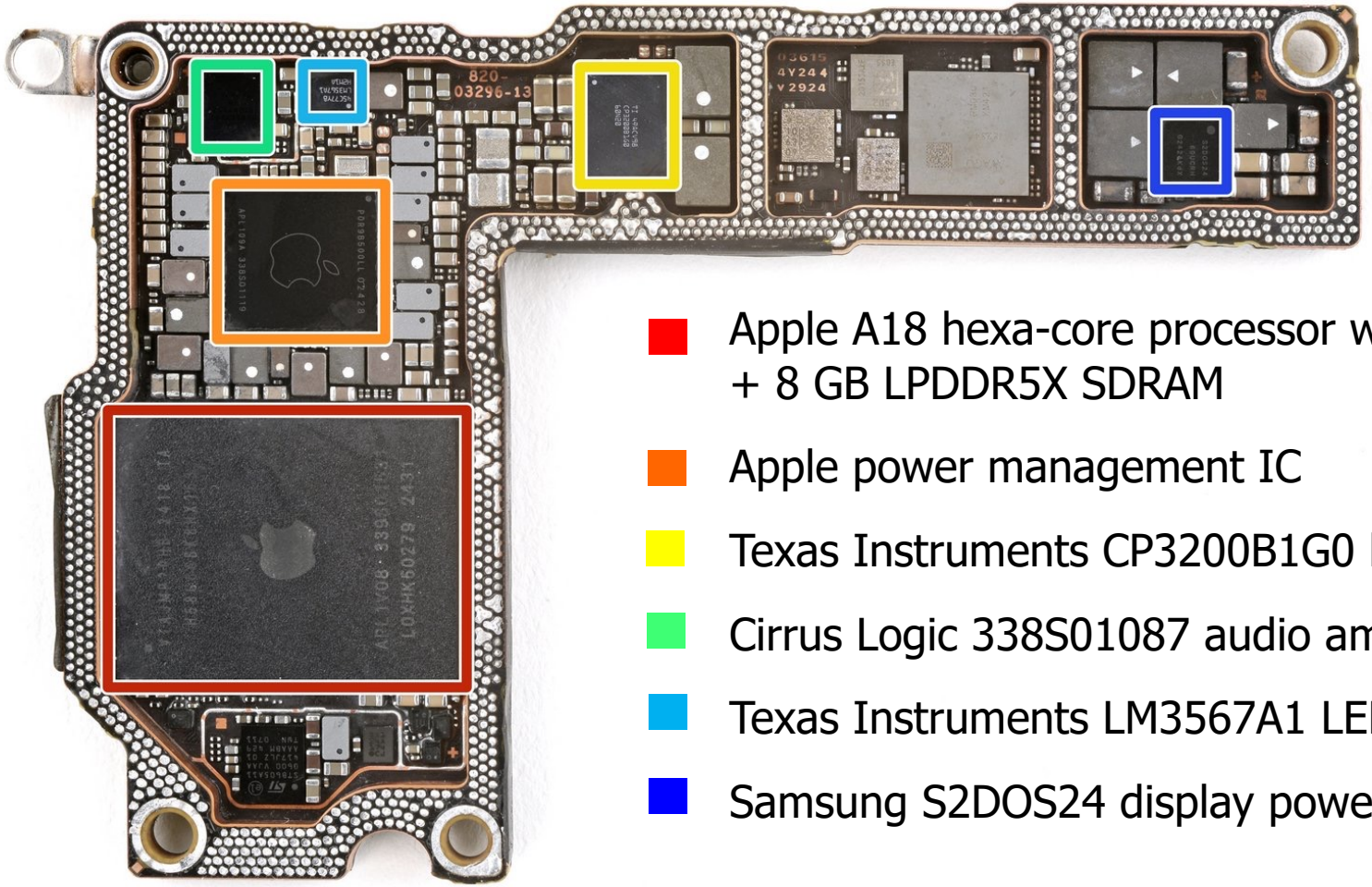
The iPhone 16's logic board contains two double-sided PCBs:

- RF board top
- RF board bottom
- Main board top
- Main board bottom

source: nl.ifixit.com

Importance of Digital Systems

- iPhone 16 Teardown: main board top

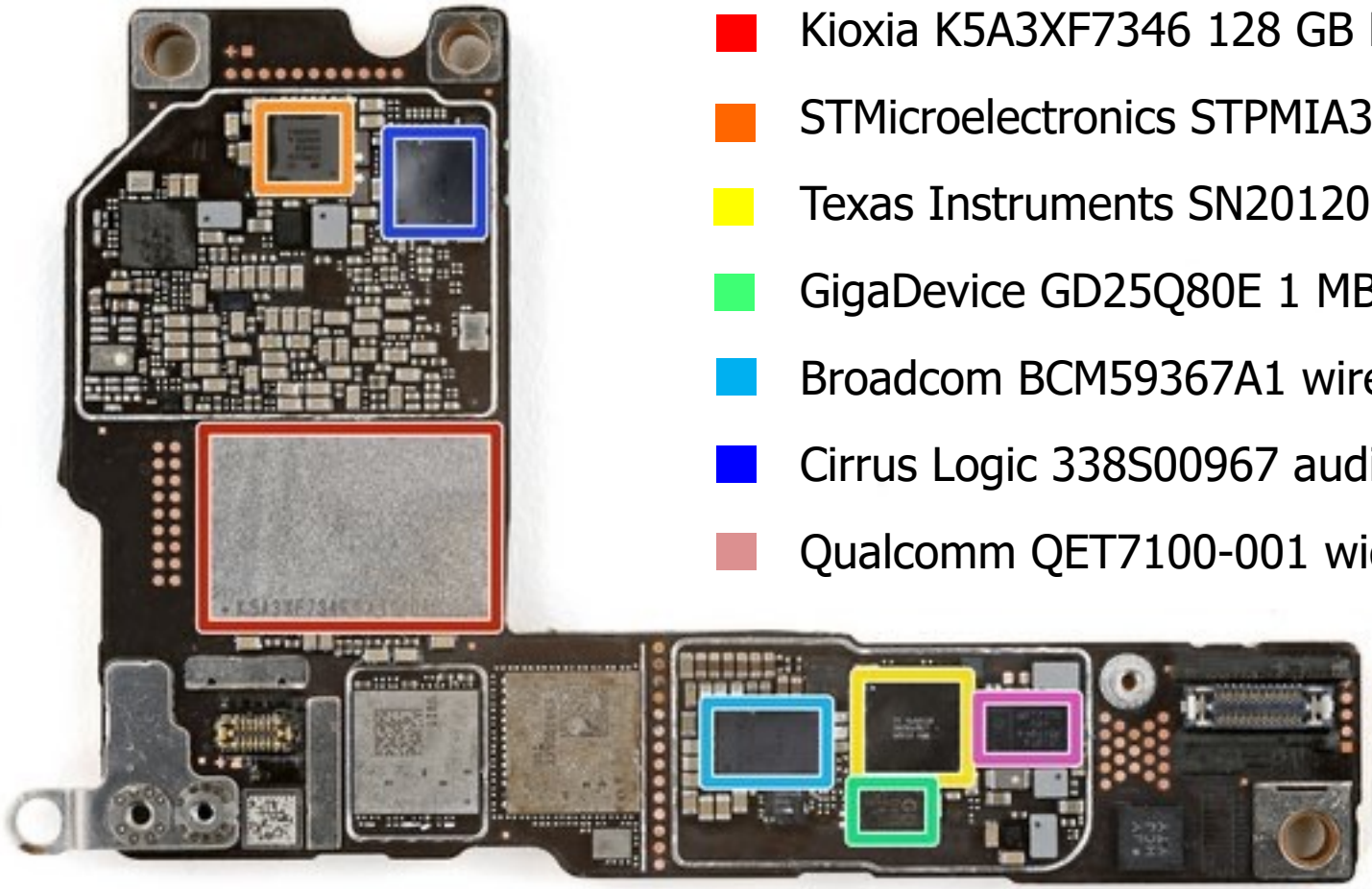


- Apple A18 hexa-core processor w/ GPU & Neural Engine ST + 8 GB LPDDR5X SDRAM
- Apple power management IC
- Texas Instruments CP3200B1G0 battery charger
- Cirrus Logic 338S01087 audio amplifier
- Texas Instruments LM3567A1 LED flash driver
- Samsung S2DOS24 display power supply

source: nl.ifixit.com

Importance of Digital Systems

- iPhone 16 Teardown: main board bottom

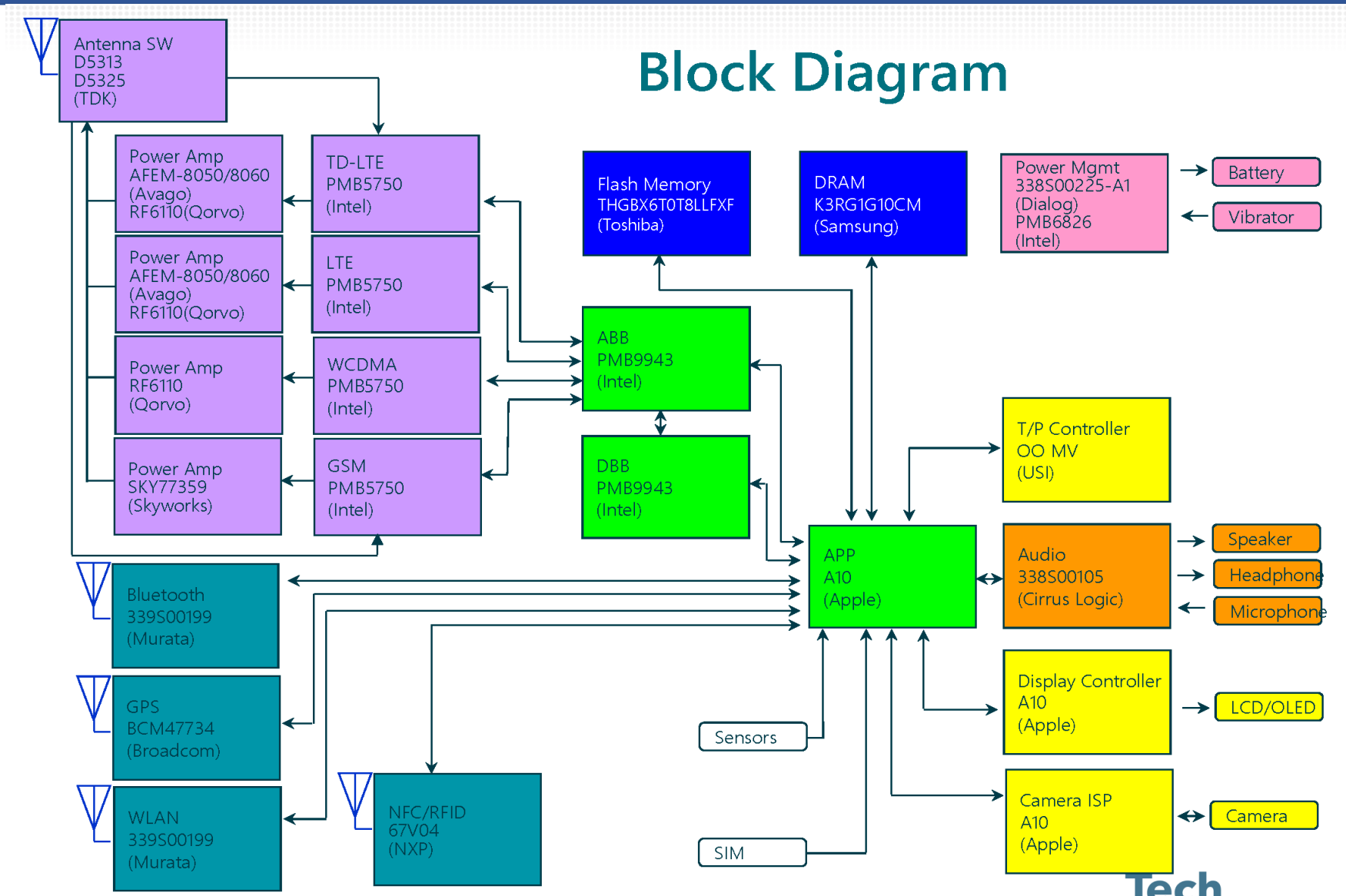


- Kioxia K5A3XF7346 128 GB NAND flash memory
- STMicroelectronics STPMIA3A426AEQ power management
- Texas Instruments SN2012027 USB type-C controller
- GigaDevice GD25Q80E 1 MB serial NOR flash memory
- Broadcom BCM59367A1 wireless charging controller
- Cirrus Logic 338S00967 audio codec
- Qualcomm QET7100-001 wideband envelope tracker

source: nl.ifixit.com

Importance of Digital Systems

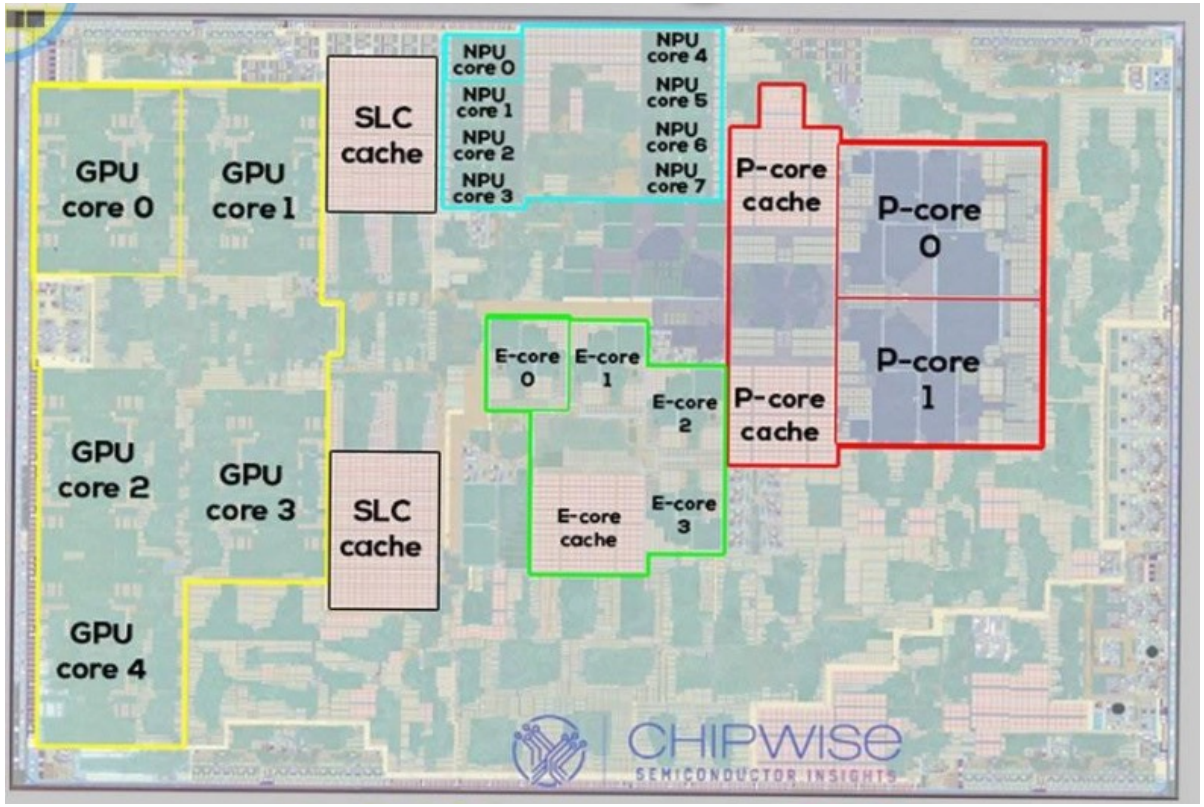
- iPhone 7



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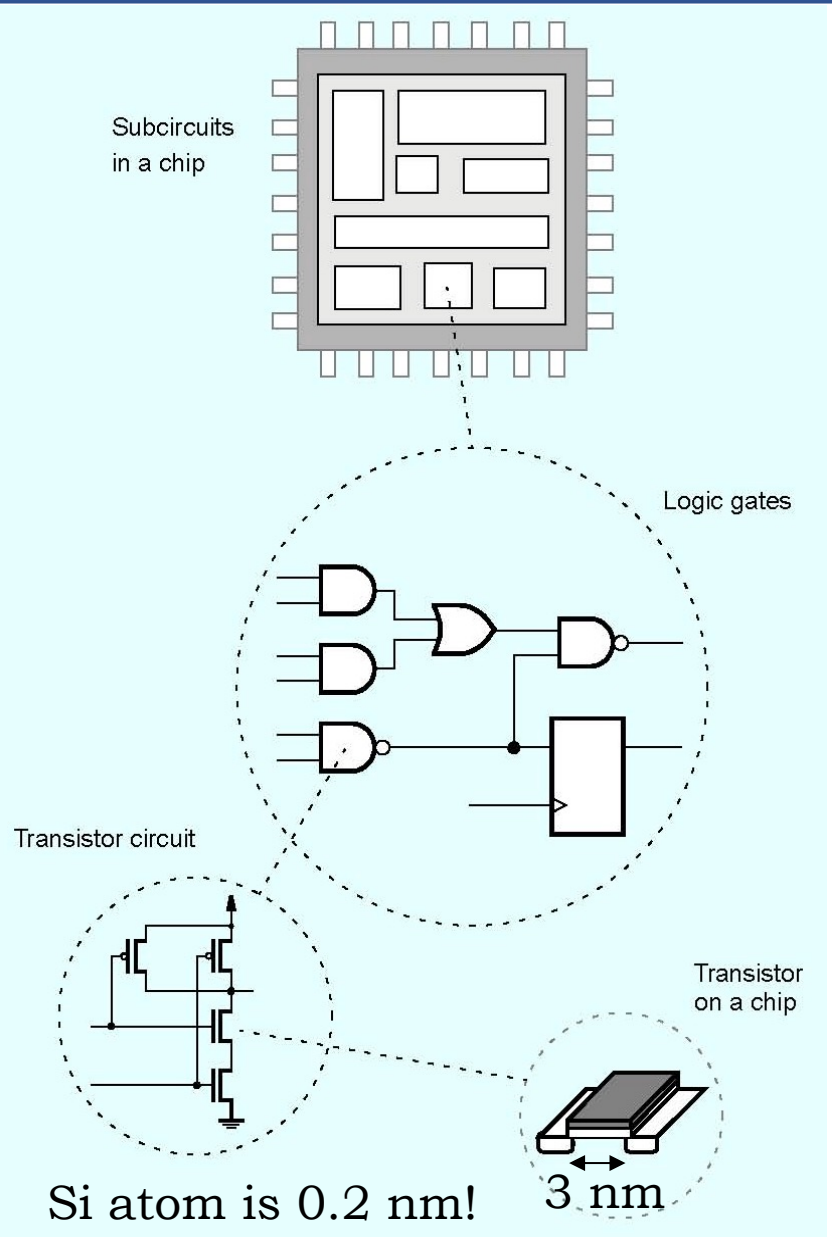
Importance of Digital Systems

- iPhone 16 A18 Chip

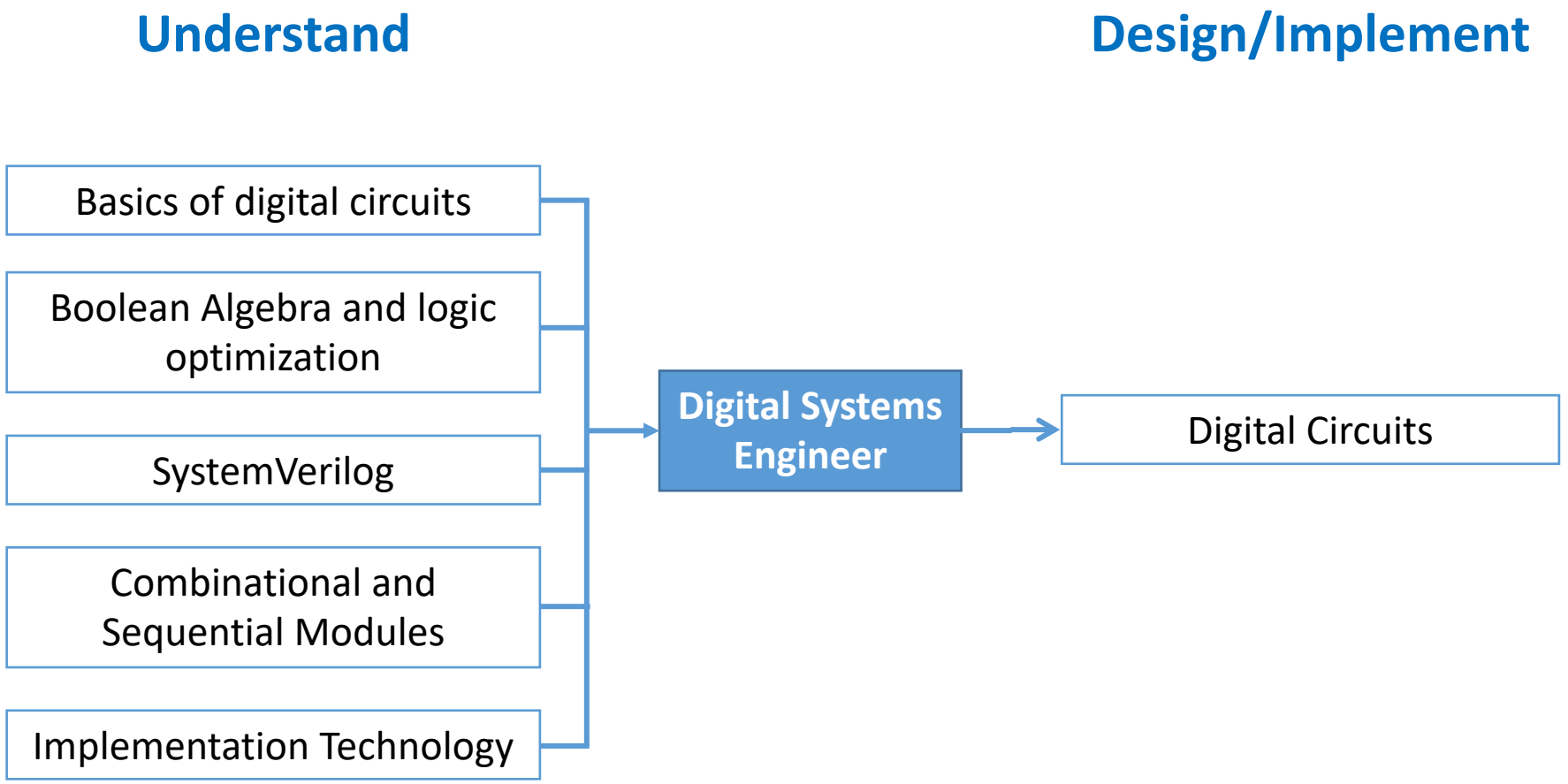


source: https://www.chipwise.com

- 3 nm technology
- 90 mm²
- 15.2 x10⁹ transistors
- 2+4 64-bit ARM proc. cores (4.05 GHz and 2.42 GHz)
- 5 GPU cores
- 16 core neural network engine

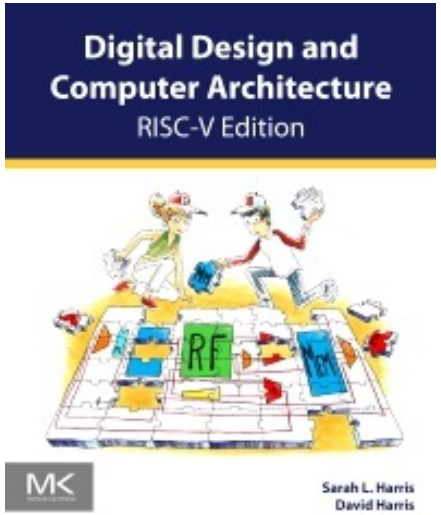


Course Overview



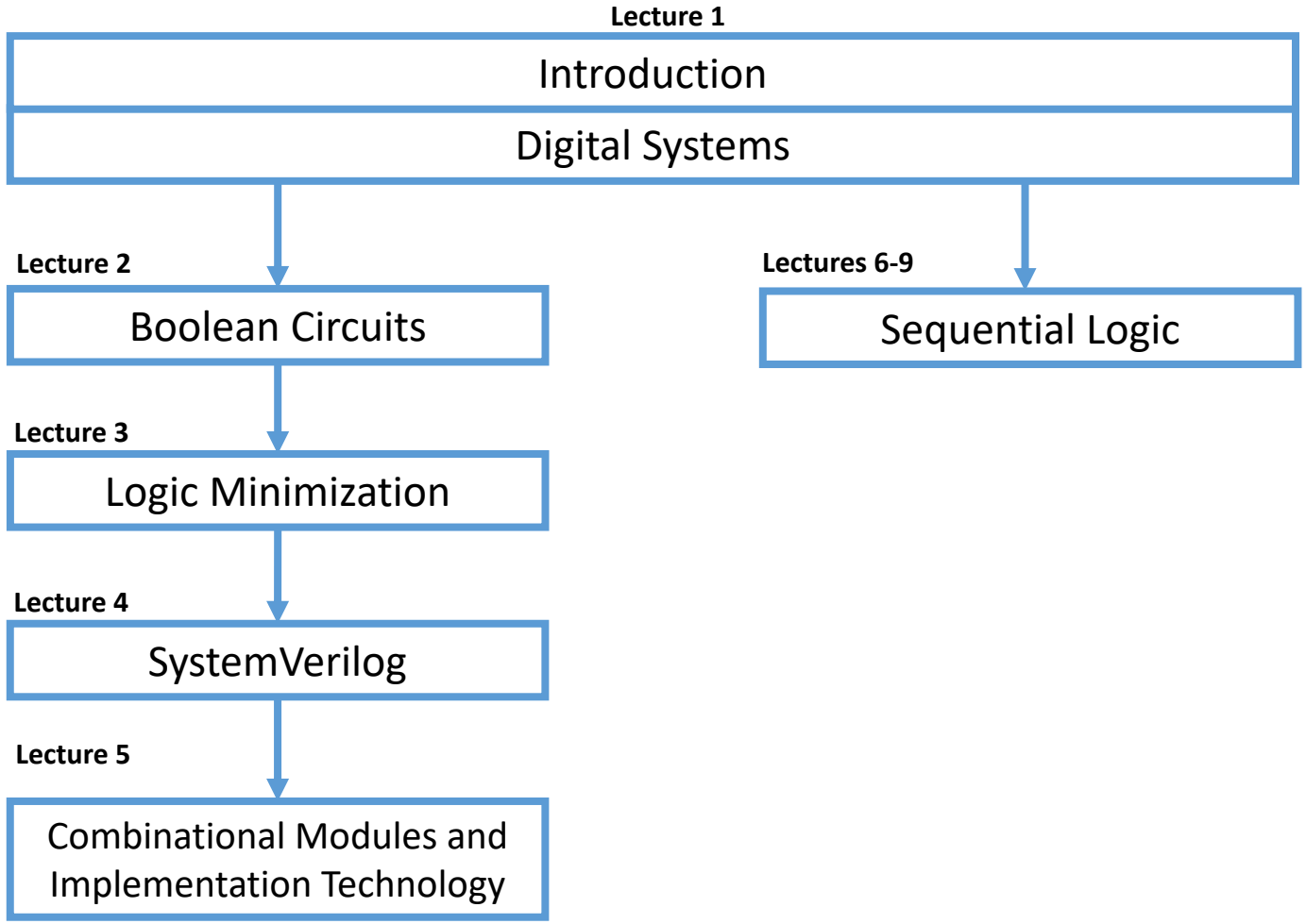
Course Overview

- Study Material
  - Lecture-slides (on Brightspace)
  - "Digital Design and Computer Architecture RISC-V edition", 2nd Edition, Sarah L. Harris, David Harris, Morgan Kaufmann, 2022, ISBN 978-0-12-820064-3 (this book will also be used for DS-B)



- Study material may be used during exams

Course Overview



- Lectures 1-5
  - 5 sets lecture assignments
  - 2 course labs
  - test 1 or trial exam (on-campus, week 4)
  - 5 seminars, including Q&A
  - 1st partial exam (week 5)
- Lectures 6-9
  - 4 sets lecture assignments
  - 2 course labs
  - test 2 (online, week 9)
  - 4 seminars, including Q&A
  - 2nd partial exam (week 10)
- Re-exam (6 Jan 2027)

Grade:

When R is average of the tests and T is average of the partial exams,
then final grade C = T + R x (10 - T) / 140; course lab sessions must be completed

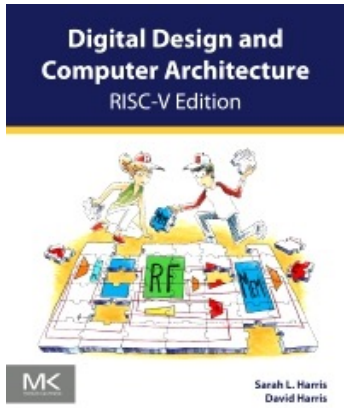
Course Overview – Schedule

week	Monday	Tuesday	Wednesday	Thursday	Friday
1		Introduction to Digital Systems			seminar 1
2	Boolean Circuits	Logic Minimization			seminar 2
3	Intro SystemVerilog	seminar 3		course lab 1	
4	Combinational Modules and Implementation Technology	seminar 4	Test 1 (trial exam) (10:45 – 11:45)	course lab 2	seminar 5
5	Exam 1 Friday Oct. 2 (9:00 – 11:00)				

- Every lecture comes with a set of lecture assignments. Finish them before the next lecture.
- On Brightspace we have described per week what you must do for the course.

Course Overview: Reading Material

Week	Date	Lecture	Topics	Study Material in book
1.1	01/09	Lec 1	Introduction to Digital System	Sections 1 – 1.5.4
1.2	07/09	Lec 2	Boolean Circuits	Sections 1.5.5, 1.5.6, 2.1, 2.3 (not yet 2.3.5) and 2.4
	08/09	Lec 3	Logic Minimization	Sections 2.2, 2.3.5, 2.5, 2.7 and 2.9 (until "If we choose to, we can avoid this glitch").
1.3	14/09	Lec 4	SystemVerilog	Sections 2.6.1, 4.1 - 4.3 (not tri-state buffer and value z) and 4.9 (not example 4.39)
1.4	21/09	Lec 5	Combinational Modules and Implementation Technology	Sections 1.6, 1.7.4 - 1.7.7 and 2.8



The lecture slides are also part of the study material!

Course Overview

- Assignments
  - Lecture assignments (via grasple.com) are available on Brightspace
    - Try to finish them before the next lecture
  - Info about course lab sessions also on Brightspace
    - Homework part has to be completed before entering the lab
    - Finishing lab assignments is compulsory

Course Overview

- Weekly Tasks
  - On Brightspace you can find the “Weekly Tasks”.
  - Per lecture the tasks are:
    a. Study the slides
    b. Read the relevant text in the book
    c. Do the assignments
  - Make sure that a and b are done before c.

Course Overview: Fraud during the Lab

- The lab is an integral part of the course! Committing fraud can have serious consequences:
  - You will be immediately expelled from the lab
  - You will have to answer to the exam committee

- For more information, see:
  - https://www.tudelft.nl/en/student/my-study-me/rules-guidelines-and-participation/fraud-plagiarism

- In case you (expect to) run into problems: contact the responsible staff immediately!

Course Overview: Learn by Practice

- You learn by looking at examples and by trying it yourself!

- Although study material may be used at the exam: practice and learn important things by heart! Otherwise:
  - You will not be able to solve the questions.
  - You will not have enough time.

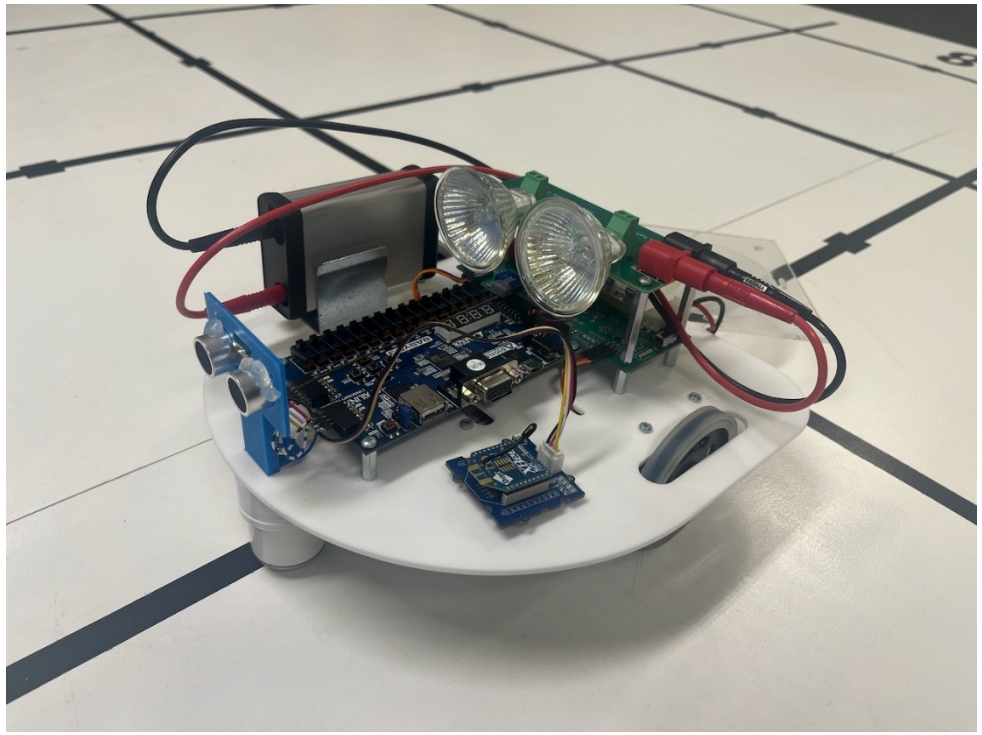
- Note that there are old example exams available on Brightspace for practising.

Course Overview: Next Courses

- Year 1
  - EE1D2 Digital Systems B (Q3)
  - IP-2 Smart Robot Challenge (Q4)
- Year 2
  - EE2C2 Mixed-Signal Circuits and Systems
  - EEX05 Chip Design (elective)
- Year 3
  - EE3D1 Computer Architecture and Organization

Course Overview

IP-2 Project "Smart Robot Challenge



Robot searching its way in a maze

- local control on FPGA: SystemVerilog hardware language
- global control on PC: C programming language
- wireless communication via ZigBee

EE1D1: Digital Systems A

BSc. EE, year 1, 2025-2026, lecture 1

Introduction to Digital Systems

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Outline

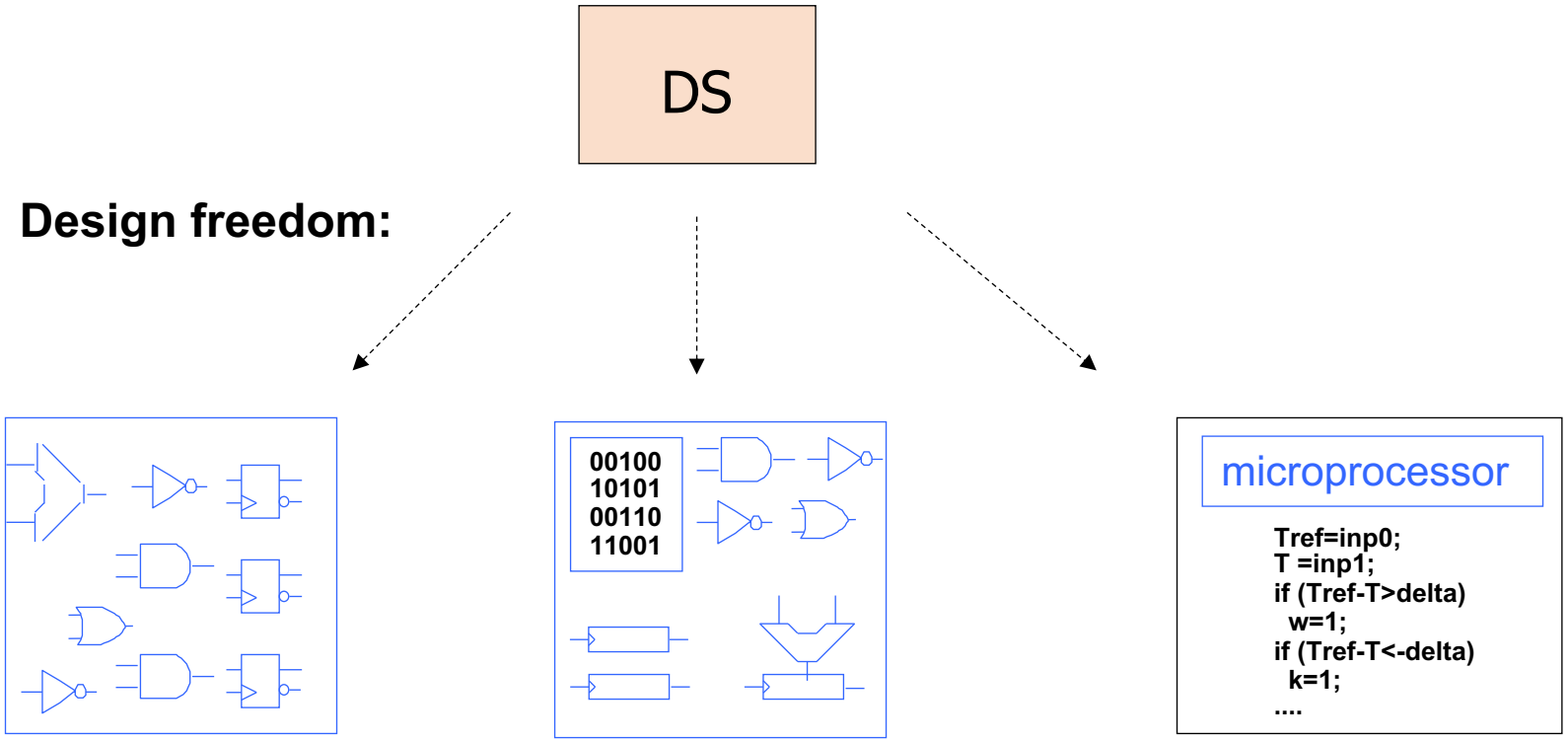
- Learning Objectives
- Digital Systems
- Digital versus Analog
- Binary Systems and Boolean Algebra
- Data Representation

Learning Objectives

As student you should be able to:

- Explain the importance of digital systems
- Describe the benefits of digital systems over analog systems
- Get familiar with binary systems and basic Boolean algebra
- Convert numbers between different base systems (e.g., from decimal to binary and vice versa)
- Compute with negative and positive numbers

Digital Systems

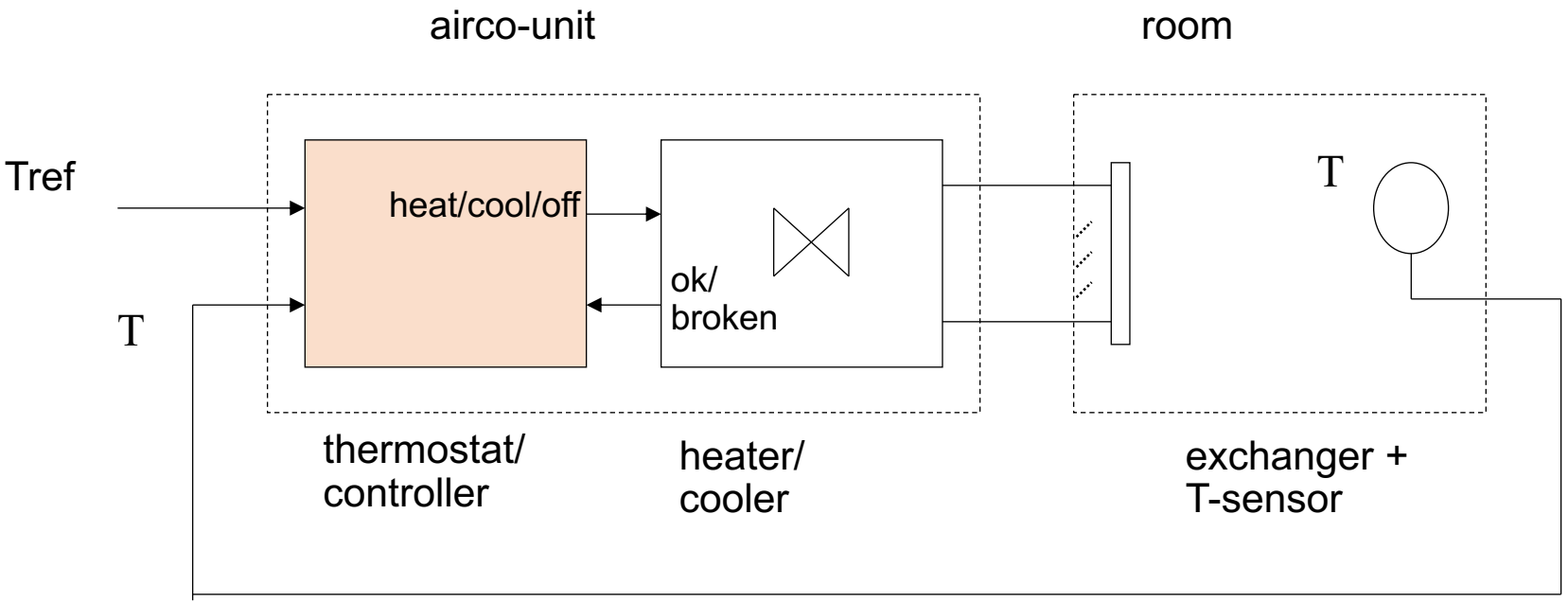


advantages: speed
parallelism

flexibility
complexity

Simple example: Airco thermostat/controller





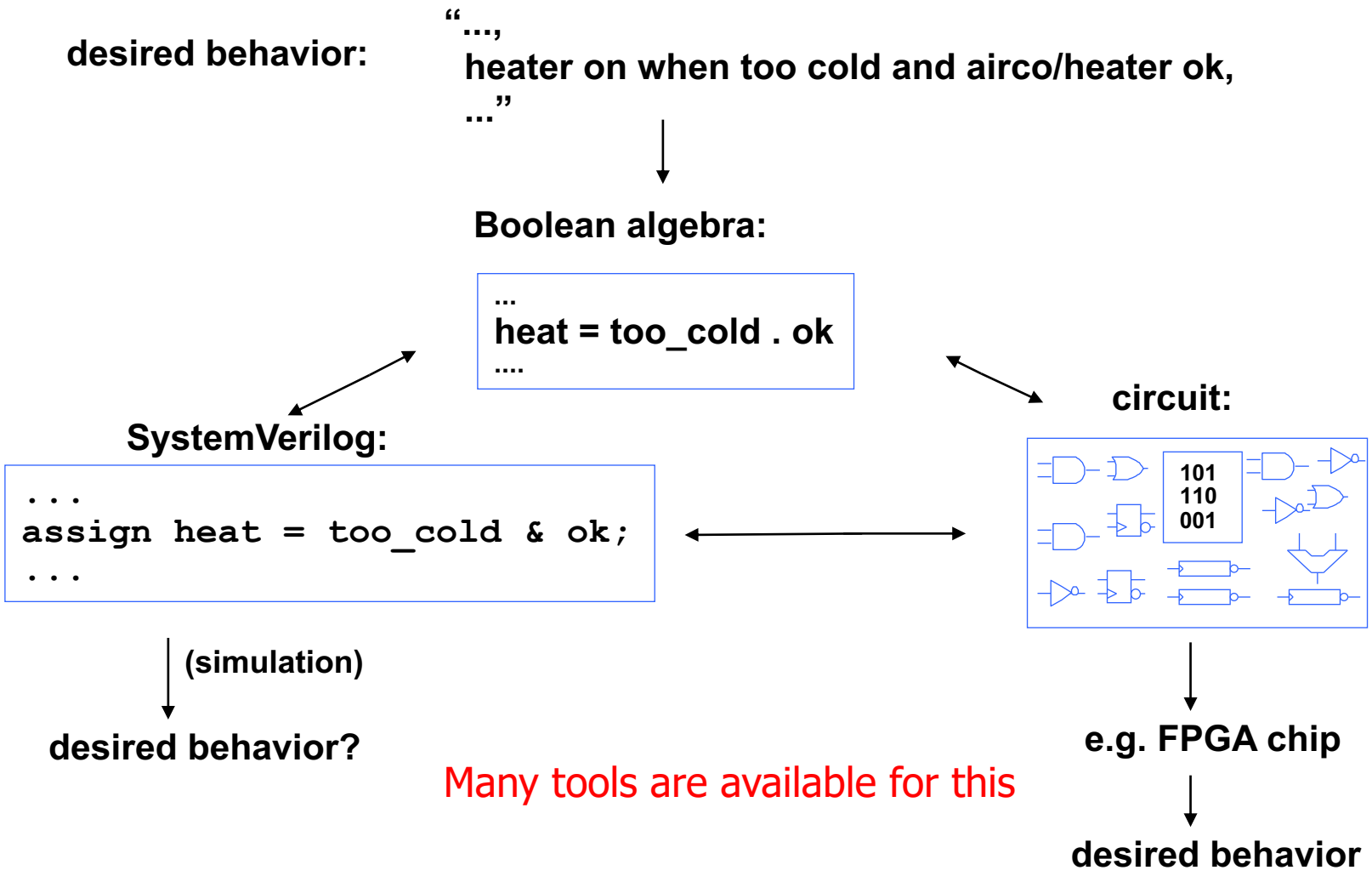
= computer system, specification =

Tref-T > delta AND ok --> heat
Tref-T < -delta AND ok --> cool
|Tref-T| <= delta OR broken --> off

compute and decide

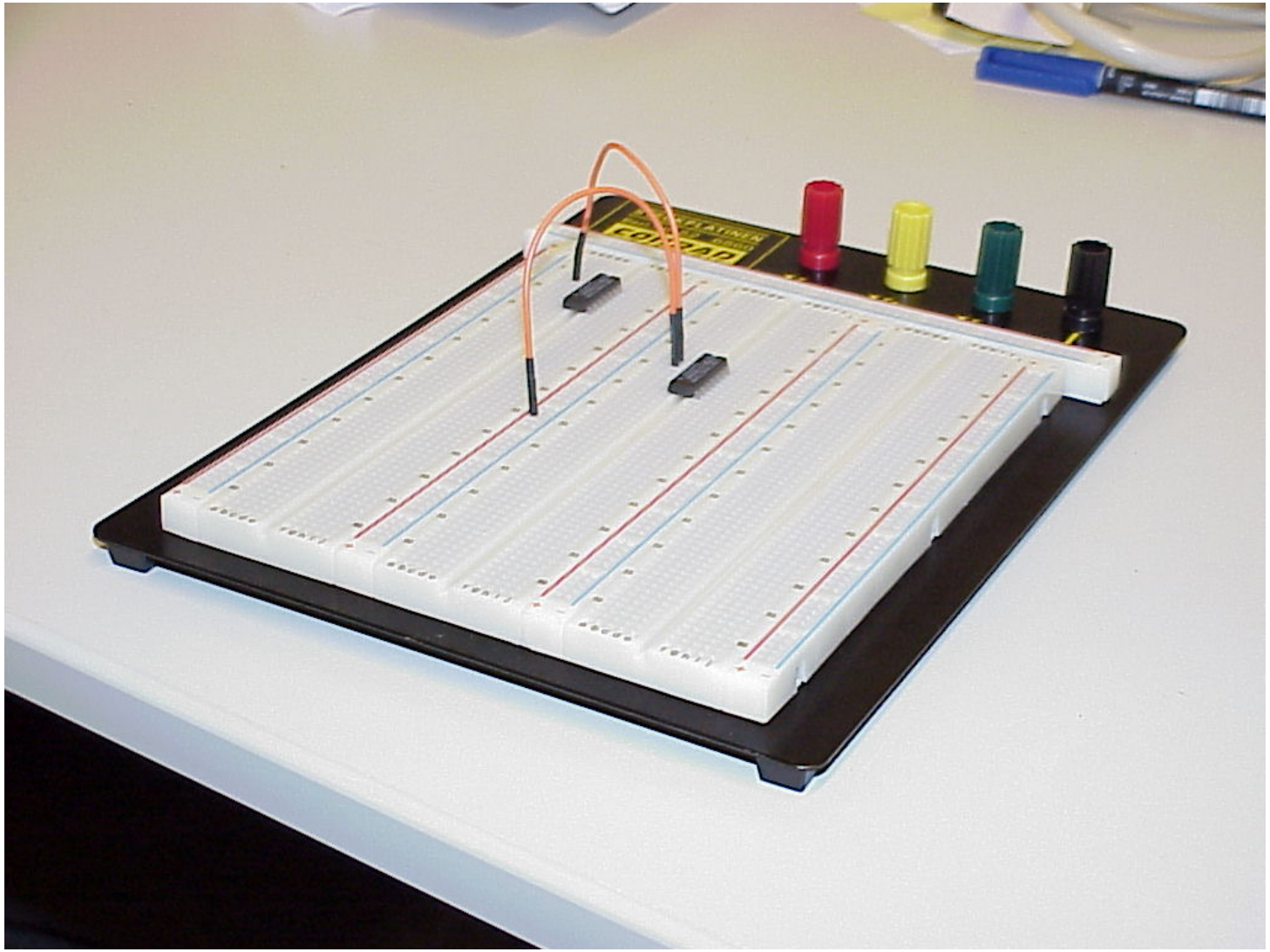
Digital Systems

Digital circuit design in practice



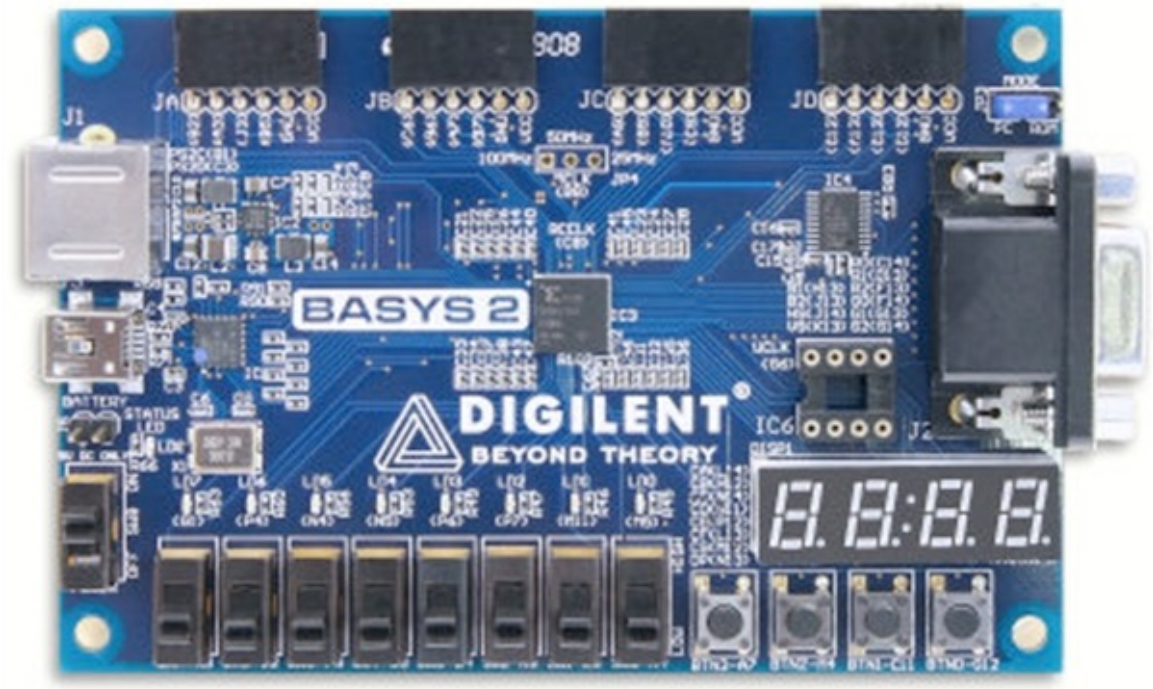
Digital Systems

Experiment hardware: Breadboard

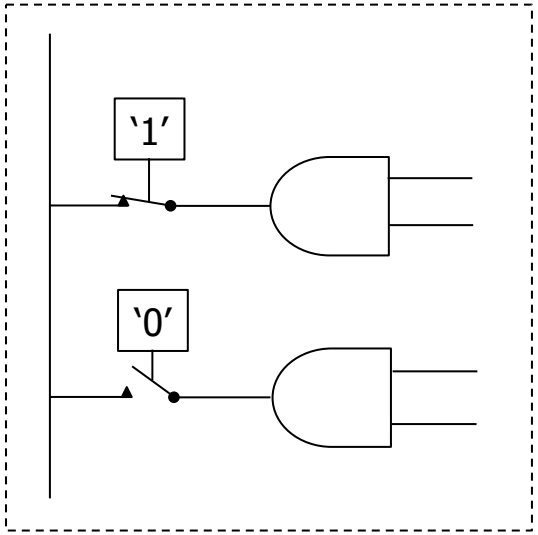


Digital Systems

BASYS2 Board with Xilinx Spartan 3E FPGA



FPGA contains components that are connected by switches controlled by RAM memory



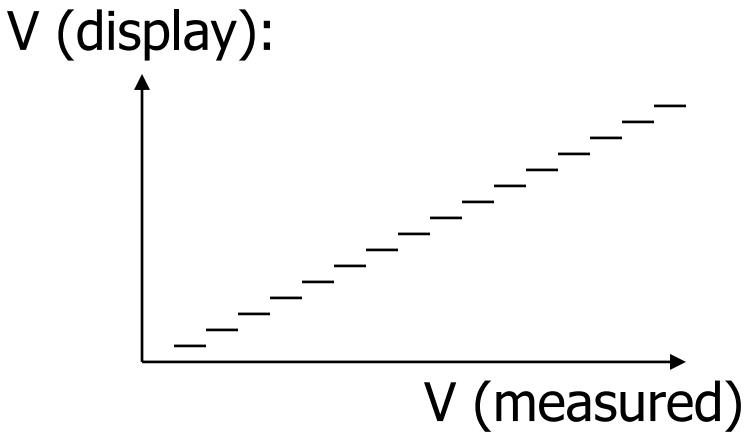
EE1D1: Digital Systems A

Digital Versus Analog

Digital versus Analog

Digital systems: represent analog values (e.g. temp, voltage, natural constants) by a *finite* number of digits

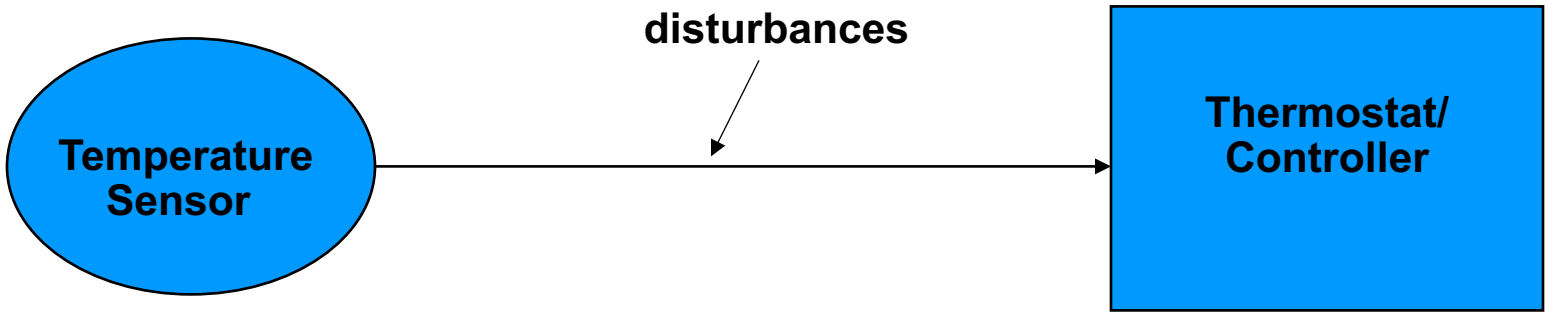
E.g. digital voltage meter:



Other examples:

- voltage meter with pointer: analog
- thermometer met digital display: digital
- mercury thermometer: analog
- π: analog
- 3.1415: digital
- e: analog
- old fashioned clock: analog

Digital versus Analog



How can we best transfer a temperature in the range 0 – 100 °C with 0.1 °C accuracy? Let's say 25.0°C

- Analog: e.g. using 1 signal wire with voltage range 0…10V:

  25.0°C = 2.50V

  accuracy has to be 10/1000=0.01 V

- Digital: e.g. using 3 signal wires, each using 10 different levels: 0V, 1V, 2V, …, 7V, 8V, 9V

  25.0°C ⇒ 2V,5V,0V = 250

  $(2 \bullet 10^2 + 5 \bullet 10^1 + 0 \bullet 10^0) \bullet 0.1°C = 25.0\ °C$

  accuracy has to be 10/10 = 1V,
  deviations of 0.1 - 0.4V are no problem

- Advantages:
  - Less sensitive to noise
  - Simpler circuits
- Disadvantage:
  - more signal wires and components (but plenty are available on a modern chip!)

EE1D1: Digital Systems A

Binary Systems and Boolean Algebra

Binary Systems and Boolean Algebra

Use variables with only two discrete values (binary values):

- 1, true, ... (e.g. 10 V, 25 mA, …)
- 0, false, ... (e.g. 0 V, 0 mA, …)

bit = **bi**nary digi**t**

Advantages:

- even less sensitive to noise
- even more simple circuits

- To represent all kind of data and perform computations on it:
  - 25.0°C ⇒ 10 V, 10 V, 10 V, 10 V, 10 V, 0 V, 10 V, 0 V = 11111010 = 250
  - $(1 \bullet 2^7 + 1 \bullet 2^6 + 1 \bullet 2^5 + 1 \bullet 2^4 + 1 \bullet 2^3 + 0 \bullet 2^2 + 1 \bullet 2^1 + 0 \bullet 2^0) \bullet 0.1°C = 25.0°C$
  - Other quantities, can be represented by (long) sequences of 0's and 1's such as sound and video

- To make decisions based on **logic**

IF	(T_{ref} - T > D) AND (airco/heating. unit ok)
THEN*	(turn on heater)

Both condition have to be true for ''heater on" to be true

THEN* = then and only then

Binary Systems and Boolean Algebra

- Algebra defines values and possible operations on values

Boolean algebra:

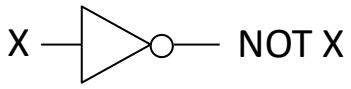
- values:
  - 0 (false), e.g. x = 0
  - 1 (true), e.g. x = 1
- operations:
  - ’, $\overline{\ }$ (NOT), e.g. f = x’ equals f = $\overline{x}$
  - • (AND), e.g. f = x • y
  - + (OR), e.g. f = x + y

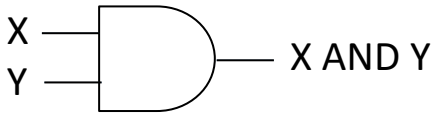
X	NOT X
0	1
1	0

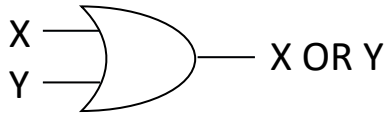
X	Y	X AND Y
0	0	0
0	1	0
1	0	0
1	1	1

X	Y	X OR Y
0	0	0
0	1	1
1	0	1
1	1	1

Symbols for use in circuit diagram:





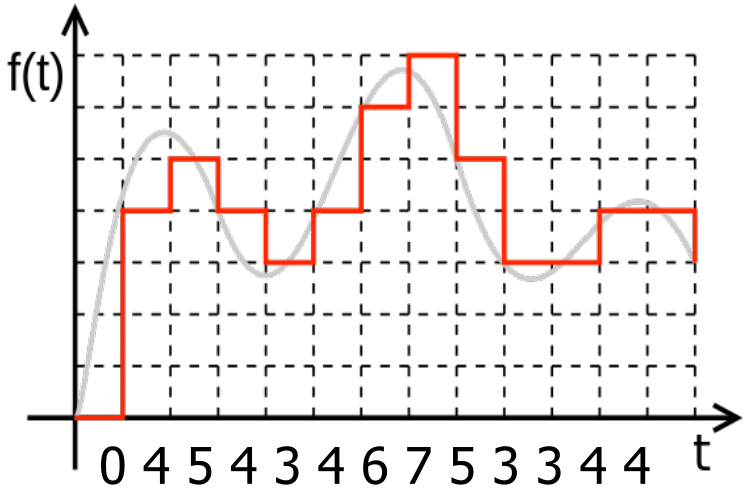


EE1D1: Digital Systems A

Data Representation

Data Representation – Approximation

- All kind of information can be approximated by sequences of numbers:



- Examples:
  - Text: characters are represented by a number
  - Sound: air pressure amplitude as function of time points t_1, t_2, t_3, ...
  - Image: light intensity for a matrix of red, green and blue (RGB) pixels
  - Video: matrix of RGB pixel values as function of time points t_1, t_2, t_3, ...

- Sometimes Giga bits or Tera bits are required (e.g. for video)

Data Representation – Natural Numbers and Characters

number	binary representation
0	00000000
1	00000001
2	00000010
3	00000011
4	00000100
5	00000101
.	. . . .
254	11111110
255	11111111

with 8 bits $2^8 = 256$ possibilities

8 bits = 1 Byte

character representation

.	. . . .
'%'	00010101
'&'	00010110
.	. . . .
'A'	01000001
'B'	01000010
'C'	01000011
'D'	01000100
.	. . . .
'a'	01100001
'b'	01100010
'c'	01100011
.	. . . .

ASCII coding

Clearly: More bits per representation, more possible values:
32 bits → $2^{32} = 4.294.967.296$ values

Data Representation – Decimal, Binary, Octal and Hexadecimal

Decimal (base 10, digits 0 ... 9)

$$26_{10} = 2 \cdot 10^1 + 6 \cdot 10^0$$

Binary (base 2, digits 0, 1)

$$11010_2 = 1 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 0 \cdot 2^0$$
$$= 26_{10}$$

Octal (base 8, digits 0 ... 7)

$$32_8 = 3 \cdot 8^1 + 2 \cdot 8^0 = 26_{10}$$

Hexadecimal (base 16, digits 0 ... 9, A, B, C, D, E, F)

$$1A_{16} = 1 \cdot 16^1 + 10 \cdot 16^0 = 26_{10}$$

Data Representation – Conversion Between Decimal and Binary

From binary to decimal:

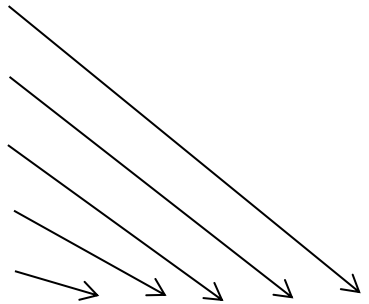
$11010 = 1 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 0 \cdot 2^0$

$= 1 \cdot 16 + 1 \cdot 8 + 0 \cdot 4 + 1 \cdot 2 + 0 \cdot 1$

$= 26$

From decimal to binary:

```
26 / 2 = 13 rest 0
13 / 2 = 6  rest 1
6  / 2 = 3  rest 0
3  / 2 = 1  rest 1
1  / 2 = 0  rest 1
```



-> 1 1 0 1 0

From decimal to binary (trial and error):

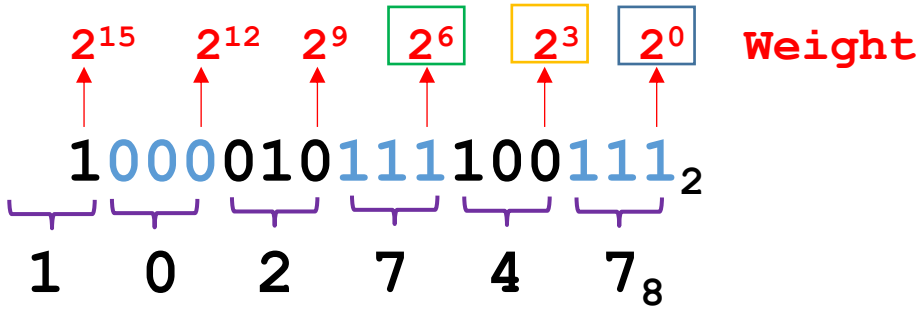
32	16	8	4	2	1	Weights
0	1	1	0	1	0	Converted value
26	10	2	2	0	0	Remainder

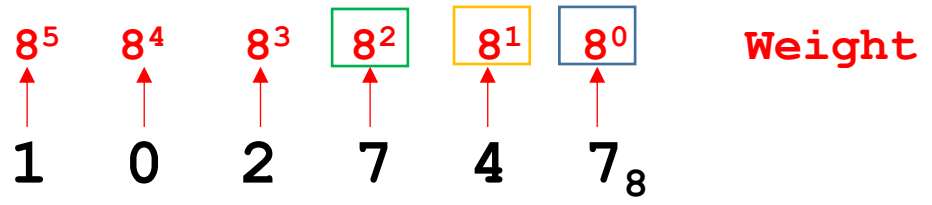
From decimal to binary (trial and error):

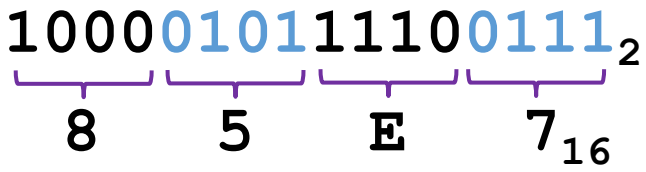
2^5 (32): 26 = 0 • 2^5 + **26** => digit 0 (MSB)
2^4 (16): 26 = 1 • 2^4 + **10** => digit 1
2^3 (8) : 10 = 1 • 2^3 + **2** => digit 1
2^2 (4) : 2 = 0 • 2^2 + **2** => digit 0
2^1 (2) : 2 = 1 • 2^1 + **0** => digit 1
2^0 (1) : 0 = 0 • 2^0 + **0** => digit 0 (LSB)

Data Representation – Binary, Octal and Hexadecimal

Simple mutual conversion:







Ideal for describing values in digital systems

Data Representation – Conversion Question:

What is the binary representation for the octal number 254_8 ?

a. 10101010
b. 10101100
c. 11010110
d. The correct answer is not listed

$254_8 = 2 \cdot 8^2 + 5 \cdot 8^1 + 4 \cdot 8^0 = 128 + 40 + 4 = 172_{10}$

172 / 2 = 86 rest 0
86 / 2 = 43 rest 0
43 / 2 = 21 rest 1
21 / 2 = 10 rest 1
10 / 2 = 5 rest 0
5 / 2 = 2 rest 1
2 / 2 = 1 rest 0
1 / 2 = 0 rest 1

=> 10101100, hence b

But easier is (in this case) per digit:
2 5 4 (octal)
= 10 101 100 (binary)

Data Representation – Positive and Negative Numbers

- Sign bit

```
5   000101
4   000100
3   000011
2   000010
1   000001
0   000000
0   100000
-1  100001
-2  100010
-3  100011
-4  100100
-5  100101
```

If 0 => positive
If 1 => negative

- Two's complement

```
 5  000101
 4  000100
 3  000011
 2  000010
 1  000001
 0  000000

-1  111111
-2  111110
-3  111101
-4  111100
-5  111011
```

-2^5 … 2^1 2^0 **Weight**

- With both representations the left-most bit represents the sign. In contrast, *unsigned* numbers don't have a sign bit.
- Two's complement appears to result in much simpler add/subtract circuits
- Trick to change sign for 2's compl: invert all bits and add 1

  3 -> 000011 -> 111100
  -> 111101 -> -3
  -3 -> 111101 -> 000010
  -> 000011 -> 3

Data Representation – Binary Computation

Addition

carry to next column →

	dec.		bin.
	1		1
	8		001000
	14	+	001110
	22		010110

Subtraction

	dec.		bin.
	22		010110
	14	−	001110
	8		001000

- But borrowing from next column is a complex operation.
- Easier is to use two's complement representation and add −14 to 22.

−14 = 110001 + 1
= 110010

		dec.		bin.
			~~1~~1 1	
		22		010110
		−14	+	110010
		8		001000

outgoing carry can be neglected in this case.

Summary

- Introduction to Digital Systems
  - Hardware
  - Software
- Digital versus Analog
  - Needs more resources
  - Digital requires less accuracy
- Binary Systems and Boolean Algebra
  - Logic/Boolean operations
- Data Representation
  - Positive/negative numbers
  - Binary, decimal, hexadecimal

To do list

- Reading material book "Digital Design":
  - Section 1 – 1.5.4

- Assignments Lecture 1



Thank you